Agent-Based Model v2.0 for Food Security Analysis

IAMO XR Lab

Leibniz Institute of Agricultural Development in Transition Economies

Theodor-Lieser Str. 2 · 06120 Halle (Saale) · Germany

Horizon Europe — SecureFood Consortium

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www.iamo.de · www.secure-food.eu

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CONSORTIUM PARTNERS

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GROCERYsim ABM v2.0 · User Manual

SECTION 1

Getting Started — Navigation

GROCERYsim ABM v2.0 uses a five-card visual navigation system. When you open the application you are presented with the Home screen displaying five large colour-coded cards. Each card expands into one or more specialised sub-sections. A fully clickable breadcrumb at the top of every sub-section (e.g. Menu › Analysis › Agent Replay) allows instant return to the Home screen or to the parent card.

1.1 The Home Screen — 5 Cards

Card 1 Data & Setup	Load the DCE participant cohort from Firebase or upload CSV/JSON. Explore demographic archetypes and the product catalogue.
Card 2 Simulation	Run the ABM: Interactive Demo with live animation, or Monte Carlo batch for statistical confidence bands.
Card 3 Analysis	Six specialised sub-sections: Food Waste, Per-Product, Behavioural Theory, Sensitivity, Compare Scenarios, Agent Replay.
Card 4 Policy & Strategy	Policy interventions, Stakeholder dashboards, Stress Test battery, Multi-Store Network, and the Finnish Regional Map.
Card 5 Export	CSV downloads for all datasets, full data bundle, and the branded 7-section PDF report.

TIP Click any sub-section button inside a card to open it. The breadcrumb is fully clickable — 'Menu' returns to the Home screen; clicking the card name shows its description.

1.2 SecureFood Scenario Simulator

The 'Scenario Simulator' button at the top right of every page opens the dedicated SecureFood climate-disruption tool for Horizon Europe partners. A 'Scenario Instructions' button beside it downloads the PDF walkthrough. The simulator runs independently of the main five-card navigation.

1.3 Onboarding Tour

Click 'Tour' in the left sidebar at any time to replay the 8-step guided tour. Navigate with Previous / Next buttons, or dismiss with 'Skip tour'.

SECTION 2

Sidebar Parameters Reference

All simulation levers are in the left sidebar. Changes take effect on the next run.

2.1 Core Simulation

Simulation Days	Number of days to simulate (default 60). Longer runs expose recovery dynamics.
Number of Agents	Consumer agents per day. Higher counts improve stability but increase run time.
Store Size	S / M / L — auto-scales shelf capacity, reorder point, and initial stock levels.
Random Seed	Fix for reproducible results; leave blank for a random seed each run.

2.2 Supply Chain

Reorder Point	Stock level (units) that triggers a replenishment order.
Lead Time (days)	Days between order placement and shelf arrival.
Order Quantity	Units ordered per replenishment cycle.
Supply Disruption	Days the supply chain is fully disrupted during the crisis window.

2.3 Crisis Settings

Crisis Start / End	Day range for the crisis shock. Baseline runs ignore this range.
Inflation Rate (%)	Price inflation applied to all products during the crisis window.
Panic Sensitivity	How strongly consumers react to low stock levels (0–1 scale).
Hoarding Multiplier	Scales how much extra stock panic-buyers attempt to purchase.

2.4 Behavioural Levers

Media Channel	Primary communication channel: social media, TV, radio, or mixed.
Purchase-Limit Nudge	Enable a per-visit purchase cap on high-demand SKUs.
Cap (units/visit)	Maximum units a consumer can buy per product per visit when nudge is active.
Stockpile Horizon	Days ahead consumers plan when hoarding behaviour activates.

TIP The Behavioural Theory sub-section in Card 3 visualises how these levers affect Prospect Theory loss aversion, TPB intention scores, and FIES food insecurity.

SECTION 3

Card 1: Data & Setup

This card is the starting point for every session and contains one sub-section: Data & Population.

3.1 Loading Data

- Firebase (live): If Firebase credentials are configured the app auto-loads the latest DCE survey cohort on startup. A spinner confirms the live connection.
- CSV / JSON upload: Use the file uploader to load an exported DCE dataset manually. Required columns: age, income, education, archetype, and product preference weights.
- Demo data: If no file is loaded the app generates a synthetic 116-respondent cohort matching the Finnish DCE study parameters.

3.2 Cohort Overview

- Demographics panel: age distribution, income brackets, education levels.
- Archetype clustering: K-Means (k = 4) segments consumers into Bargain Hunter, Quality Seeker, Health Conscious, and Convenience Shopper.
- DCE preference weights: mean willingness-to-pay for Finnish origin, organic, low-fat, and packaging attributes from the survey.

3.3 Product Catalogue

The catalogue panel displays all SKUs with categories (Dairy / Meat / Produce / Packaged), shelf life, base price, and current stock levels. It drives the per-product analysis in Card 3 and the SecureFood dairy scenario.

TIP Load or verify data before running any simulation. The demo cohort is used automatically if no file is uploaded, so you can explore the tool immediately.

SECTION 4

Card 2: Simulation

Card 2 contains two sub-sections: Interactive Demo and Scientific Analysis.

4.1 Interactive Demo

- Runs a paired Baseline vs Crisis simulation using your current sidebar parameters.
- Day-by-day animation: watch revenue, shelf stock, and consumer panic evolve in real time.
- Animation speed is adjustable via the sidebar slider (seconds per day).
- Metrics panels update each day: daily revenue, units sold, stockout events, waste, lost sales.
- Save scenario: assign a name to the current run; it is stored for later comparison in Card 3.
- Both Baseline and Crisis models share the same random seed for a controlled paired comparison.

TIP Run the Interactive Demo first after changing parameters — it populates data required by Food Waste, Per-Product, Behavioural Theory, and Agent Replay in Card 3.

4.2 Scientific Analysis (Monte Carlo)

- Executes N independent replications of the paired simulation (default N = 20).
- Displays p10 / IQR (p25–p75) / p90 percentile confidence bands on revenue charts.
- Baseline vs Crisis dual-band chart directly overlays both scenarios' uncertainty ranges.
- AI storage recommendations interpret simulation results in plain language.
- Statistical comparison table reports mean revenue, waste, and lost sales for both scenarios.
- The branded PDF report is generated from the Export card once a run is complete.

SECTION 5

Card 3: Analysis

Card 3 provides six specialised sub-sections for deep-dive analysis. All require a completed Interactive Demo run. Navigate to each via the card's sub-section buttons on the Home screen.

5.1 Food Waste

- Daily waste log: units expired per product per day for both scenarios.
- Waste drivers bar chart: identifies which SKUs generate the most waste.
- CO2 footprint: converts wasted units to kg CO2-equivalent using product emission factors.
- Overlaid Baseline vs Crisis waste trajectories on a shared axis.

5.2 Per-Product Analysis

- Select any SKU from the dropdown to view its stock, sales, and price over time.
- Stockout frequency chart: number of days each SKU registered zero stock.
- Price evolution: shows how crisis inflation affects individual product prices day by day.

5.3 Behavioural Theory

- Prospect Theory: loss aversion coefficient over time — peaks during the crisis window.
- TPB intention score: Theory of Planned Behaviour composite (PBC + Subjective Norm).
- FIES scale: Food Insecurity Experience Scale index by income bracket (FAO 2016).
- Stockpile pressure: aggregate consumer hoarding pressure index by simulation day.
- Use the Behavioural Levers sidebar to observe how media channel and nudges shift outcomes.

5.4 Sensitivity Analysis

- Parameter sweep: automatically varies each sidebar lever over a predefined range.
- Tornado chart: ranks parameters by their impact on mean revenue, waste, and lost sales.
- Interaction plot: visualises how pairs of parameters jointly affect simulation outcomes.

5.5 Compare Scenarios

- Side-by-side charts for any two scenarios saved from the Interactive Demo.
- Revenue, waste, lost sales, and panic level overlaid on shared time axes.
- Saved scenarios persist within the session; use Card 5 — Export to download them as CSV.

5.6 Agent Replay Viewer

- Day slider: step through the simulation chronologically after a completed demo run.
- Scatter plot: every shopper's Panic Level (x) vs Perceived Behavioural Control (y). Colour denotes archetype. The red dashed quadrant marks the high-panic / low-PBC danger zone.
- Archetype breakdown bar chart: daily composition of each consumer segment.
- Income vulnerability heatmap: fulfilment rate by income bracket and simulation day.

> **TIP** Run the Interactive Demo, then open Agent Replay. Use the day slider to pinpoint exactly when panic peaks and which archetypes are most severely affected.

SECTION 6

Card 4: Policy & Strategy

Card 4 offers five sub-sections for intervention design, system stress-testing, and geospatial network analysis. Each sub-section can be accessed independently from the Home screen.

6.1 Policy Analysis

- Fat tax: apply a percentage tax on high-fat products and measure consequent demand shift.
- Domestic / organic subsidy: reduce price for Finnish-origin or organic SKUs.
- Purchase-cap nudge: enforce a per-visit purchase limit on hoarding-prone products.
- Nutritional labelling: toggle front-of-pack labelling effect on health-conscious consumers.
- Each policy run produces a branded PDF policy brief downloadable from this sub-section.

6.2 Stakeholder View

- KPI dashboard: large metric cards for revenue change %, food security index, waste reduction.
- Policy narrative: plain-language summary suitable for executive and decision-maker briefings.
- Toggle between Supply Chain Actor and Policy Maker perspectives.

6.3 Stress Test

- Automated six-scenario battery: Supply Chain Collapse, Price Spike (+40 %), Panic Wave, Import Crisis, Demand Surge, and Cold-Chain Failure.
- Each scenario runs a full ABM simulation and records peak revenue loss, maximum panic level, and recovery time (days to return within 5 % of baseline).
- Results table ranks scenarios by severity and is sortable by any column.
- Identifies which shock typology poses the greatest operational risk to your store.

6.4 Multi-Store Network Simulation

- Configure up to 8 Finnish stores in an editable table: name, region, size, proximity to neighbouring stores, and hub-store designation.
- Panic contagion: elevated panic in one store decays with inter-store distance and propagates to neighbouring stores.
- Emergency redistribution: hub stores transfer surplus stock to depleted stores when stock falls below the redistribution threshold.
- Daily revenue by store is displayed with a vertical legend (supports up to 8 stores without label overlap).
- Per-store summary table: risk level, revenue loss %, and peak panic for each store.
- Click 'View Results on Regional Map' to overlay risk classifications on the Finnish map.

6.5 Regional Map

- Interactive scatter map of 33 Finnish supermarket locations across six retail chains (S-ryhmä, K-ryhmä, Lidl, Aldi, Tokmanni, M-ketju).
- Regional statistics table: store density, estimated market share, and food-security indicators by Finnish region.
- After a Multi-Store simulation run, the map displays a risk-level overlay: Critical (red) / High (orange) / Moderate (yellow) / Low (green) — matched by store name.
- A 'Re-run Multi-Store Simulation' button returns directly to the network tool.

TIP Run the Multi-Store Network simulation first, then click 'View Results on Regional Map' to identify which Finnish regions are most exposed under your crisis scenario. The two sub-sections are fully interlinked.

SECTION 7

Card 5: Export & PDF Report

Card 5 centralises all download options. A completed Interactive Demo or Scientific Analysis run is required before data becomes available.

7.1 CSV Downloads

Daily Aggregates	Revenue, waste, lost sales, stock, panic — one row per day per scenario.
Per-Product Stock	Units on shelf per SKU per day for both scenarios.
SCM Log	Supply-chain events: orders placed, deliveries received, stockout flags.
Food Waste Log	Units wasted per SKU per day with CO2-equivalent estimate.
Policy Run	Results from any policy intervention applied in Card 4.
Full Bundle	All above datasets merged into a single multi-sheet CSV file.

7.2 Generate PDF Report

Click 'Generate PDF Report' to produce a branded seven-section document:

- 1. Executive Summary — revenue contraction %, waste increase %.
- 2. Simulation Parameters — complete sidebar settings at the time of the run.
- 3. Revenue Analysis — cumulative and daily revenue charts.
- 4. Supply Chain Summary — stockout events and replenishment cycles.
- 5. Policy KPIs — included if any policy intervention was active during the run.
- 6. Stress Test Ranking — included if a stress test was completed in the session.
- 7. Methodology Note — model architecture and primary data sources.

TIP The PDF report embeds Matplotlib charts directly — no internet connection required. Run Scientific Analysis first to include confidence bands in the revenue charts.

SECTION 8

SecureFood Scenario Simulator

Scenario: Climate-Driven Dairy Supply Chain Disruption

Increasingly frequent and intense weather events, driven by climate change, are creating critical vulnerabilities within the agricultural sector, specifically threatening the stability of the milk and dairy supply chain. Extreme weather disrupts livestock feed production, driving up operational costs and causing critical stock shortages at the retail level.

The SecureFood Scenario Simulator is a dedicated tool for Horizon Europe SecureFood project partners (Grant No. 101136583). It models the impact of climate-driven disruptions on Finnish dairy supply chains and consumer food security. Access it via the 'Scenario Simulator' button at the top right of any page.

8.1 Perspective Selection

Supply Chain Actor	Focus on revenue, stockout losses, shelf availability, and replenishment for producers, distributors, and retailers.
Policy Maker	Focus on food insecurity (FIES), budget exhaustion, and equity indicators for national and regional policy design.

8.2 Scenario Configuration

Disruption Type	Drought / Flood / Heat Wave / Cold Snap — each affects domestic dairy yield differently.
Duration (days)	Length of the climate disruption event in simulation days.
Severity (0–1)	Proportional reduction in domestic dairy supply caused by the event.
Import Dependency	Share of dairy demand met by imports — determines exposure to domestic disruption.
Crisis Start Day	Day within the simulation window on which the disruption begins.

8.3 Output Charts (9 Panels)

- Inflation-Volume Decomposition: constant-price vs nominal revenue over time.
- Stockout Revenue Loss: daily and cumulative lost sales from empty shelves.
- Shelf Stock by Category: Baseline (dashed) vs Crisis (solid) for each dairy sub-type.
- Consumer Panic & Stockpile Pressure: panic level and hoarding index over time.
- Basket Fulfilment Rate by Income Group: % of shopping basket successfully purchased.
- FIES Severe Food Insecurity: FAO scale index by income bracket.
- Budget Exhaustion & Gini Access Index: inequality in food access during the crisis.
- Domestic vs Import Sales Volume: food sovereignty and supply-chain resilience indicator.
- Policy Scenario Comparison: side-by-side effect of three intervention options.

TIP Each chart includes an expandable analysis box with quantified findings and peer-reviewed citations. A full scenario walkthrough is available as a downloadable PDF via the 'Scenario Instructions' button on the main page.

SECTION 9

Model Architecture & Technical Notes

9.1 Computational Framework

- Built on Mesa (Python ABM framework) — each simulation step corresponds to one shopping day.
- Consumer agents: a fresh ConsumerAgent object is created for each shopper on each day and removed at the end of that day.
- Product agents: persistent ProductAgent objects manage stock levels, order cycles, price evolution, and waste accrual.
- Scheduler: RandomActivation — agents execute in a randomised order each step to avoid artefactual ordering effects.

9.2 Consumer Decision Model

- Theory of Planned Behaviour (TPB): attitude, subjective norm, and Perceived Behavioural Control (PBC) jointly determine purchase intention for each agent each day.
- Prospect Theory loss aversion: consumers systematically over-weight the risk of stockouts, amplifying panic-buying behaviour. Default lambda = 2.25 (Kahneman & Tversky, 1979).
- DCE-derived preference weights: willingness-to-pay for Finnish origin, organic, low-fat, and packaging attributes calibrated from N = 116 Finnish consumers (Horizon Europe SecureFood survey, 2024–2025).
- Income brackets: Low (< €1,500/mo), Mid (€1,500–€3,500), High (> €3,500) — determine budget constraints and price sensitivity.

9.3 Supply Chain Model

- (s, Q) inventory policy: an order of Q units is placed whenever stock falls below the reorder point s.
- Lead time: replenishment orders arrive after a user-configurable number of days.
- Perishable waste: units exceeding shelf-life are removed from stock and logged.
- Multi-store extension: panic contagion uses a proximity-decayed transfer function; hub stores buffer emergency redistribution to depleted stores.

9.4 Data Sources

DCE Survey	116 Finnish respondents; willingness-to-pay elicited via stated preference methods (2024–2025).
Store Locations	33 Finnish supermarkets georeferenced via OpenStreetMap (2025).
FIES	FAO Food Insecurity Experience Scale methodology (FAO, 2016).
Emission Factors	Product-level CO2-equivalent estimates from the European Environment Agency (2023).
Prospect Theory	Kahneman, D. & Tversky, A. (1979). Econometrica, 47(2), 263–291.

SECTION 10

Glossary & Citation

10.1 Glossary of Key Terms

ABM	Agent-Based Model — a simulation in which individual agents follow rules and interact to produce emergent system behaviour.
Archetype	Consumer segment: Bargain Hunter, Quality Seeker, Health Conscious, or Convenience Shopper.
Crisis Period	The day range during which supply disruption and/or inflation shocks are active in the model.
DCE	Discrete Choice Experiment — a survey instrument used to elicit consumer preferences via hypothetical choice tasks.
FIES	Food Insecurity Experience Scale (FAO) — an 8-question ordinal scale measuring the severity of food insecurity.
Fulfilment Rate	The share of a consumer's intended shopping basket that was successfully purchased in a single visit.
Gini Access Index	An inequality measure applied to food access; 0 = perfect equality, 1 = maximum inequality.
Hoarding Multiplier	A parameter scaling the additional stock that panic-buyers attempt to purchase above their normal basket.
Lead Time	The number of days between the placement of a replenishment order and its arrival on shelf.
Lost Sales	Revenue foregone because a product was out of stock at the time of intended purchase.
Monte Carlo	A statistical method in which the simulation is run many times with different random seeds to produce percentile bands rather than point estimates.
Panic Level	An agent-level indicator (0–1) reflecting perceived stockout risk and urgency to hoard.
PBC	Perceived Behavioural Control — the TPB construct measuring the perceived ease or difficulty of performing an action.
Reorder Point	The stock level (in units) that triggers the automatic placement of a new supply order.
SKU	Stock Keeping Unit — a unique identifier for a specific product variant in the catalogue.
TPB	Theory of Planned Behaviour — attitude + subjective norm + PBC jointly determine behavioural intention.
Waste	Units removed from the model because they have exceeded their maximum shelf-life date.

10.2 Citation

How to Cite This Software

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Software available at: https://github.com/IvanDuric/Finland_ABM